

Heterogeneous-scale figure preview

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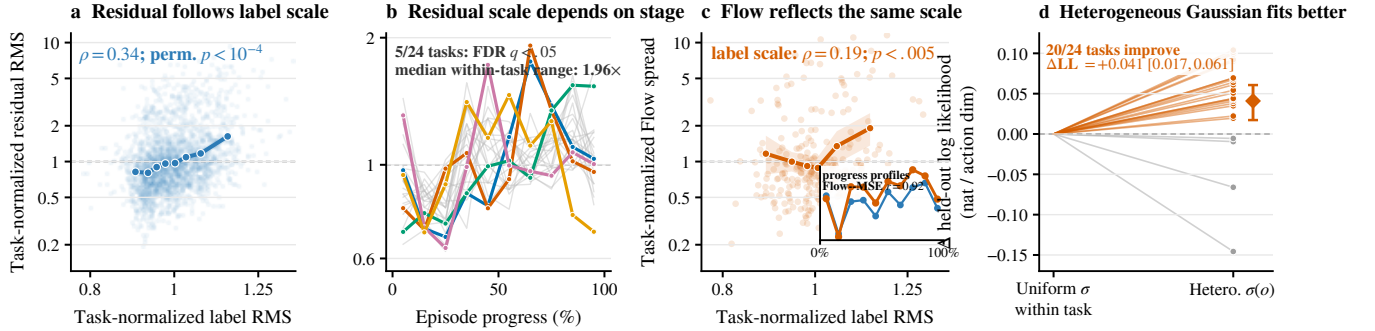


Figure 1: Robot-policy residuals have input-dependent scale. We analyze the continuous action channels of GR00T N1.7 on 24 RoboCasa-GR1 tasks. (a) Across 2,880 policy-held-out states, the RMS residual of a converged MSE policy increases with the action-label RMS after within-task normalization ($\rho = 0.34$; within-task permutation $p < 10^{-4}$). (b) Each curve traces residual scale over progress within one task. The five colored tasks retain a stage effect after Benjamini–Hochberg correction ($q < .05$); the other 19 tasks are shown in gray. The median within-task max/min scale ratio is $1.96\times$. (c) At 240 matched inputs, we sample 32 action chunks from Flow and define spread as their RMS deviation from the sample mean. Flow spread correlates with label scale ($\rho = 0.19$, $p < .005$), and its progress profile closely tracks the MSE residual profile ($r = 0.92$). (d) Keeping the action mean fixed, a cross-fitted Gaussian with input-dependent scale attains higher held-out log likelihood than one with uniform scale within each task on 20 of 24 tasks ($+0.041$ nat per action dimension; 95% task-bootstrap CI [0.017, 0.061]). Together, these results motivate heteroscedastic regression.